

WATSON AGBRAMU

Senior Software Engineer

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[LinkedIn](#) ♦ [Portfolio](#) ♦ [GitHub](#)

Ontario, Canada

SUMMARY

Senior Software Engineer with 6+ years building LLM-powered products, multi-tenant platforms, and distributed systems across SaaS, media, and fintech. Ships end to end: production RAG and agentic systems, high-throughput data pipelines, and the client-facing APIs and interfaces on top of them. Owns architecture from design through delivery.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Rust
AI Engineering	LangChain, LangGraph, OpenAI SDK, Anthropic SDK, RAG, vector search, embeddings, agentic workflows, tool calling, evals
Backend	FastAPI, Flask, Node.js, Express, GraphQL, REST, Celery, event-driven architecture
Frontend	React, Next.js, React Native, Tailwind CSS
Data Engineering	DBT, Snowflake, Airbyte, Azure Databricks, Azure Data Factory
Databases	PostgreSQL, MongoDB, MySQL, Redis, Firestore, vector databases
Cloud	AWS (Lambda, SQS, S3, CloudFront, Route 53, SAM), GCP, Azure
DevOps	Docker, Terraform, ArgoCD, Argo Workflows, GitHub Actions, Jenkins
Practices	System Design, Distributed Systems, Multi-tenant Architecture, Agile

EXPERIENCE

Senior Full Stack Developer	Sept 2024 – Present
LipDub AI (MARZ)	<i>Remote</i>

- Initiated and designed the personalization epic, delivering the API in production use today that regenerates voice and lip-sync for edited text segments, eliminating full-asset re-renders on client revisions
- Built and own the public video-generation API and the internal service layer consumed by the Next.js frontend using Python, Flask, and FastAPI, powering client integrations across generation, asset management, and account workflows
- Architected the platform’s feature-gating system using Stripe entitlements as the plan-level source of truth with LaunchDarkly as a per-tenant resolver, enabling staged rollout and tenant-level overrides without building a bespoke entitlements service, and kept gate checks off the critical path with short-TTL caching invalidated on Stripe plan-change webhooks
- Shipped custom subdomain support for enterprise clients on AWS CloudFront and Route 53, automating CNAME delegation and TLS certificate provisioning while scoping edge routing to expose only client-facing watch routes rather than the full application
- Built a media ingestion worker accepting public and YouTube URLs and introduced workflow prioritization for Argo generation jobs, authoring the workflow configuration and shifting download, format handling, and validation server-side
- Delivered core product features across Python services and Next.js including voice cloning, occlusion handling, AI avatars, translation memory, and the assets library, working within a Terraform and ArgoCD deployment pipeline
- Partnered with the sales lead in client discovery sessions, translating customer pain points into scoped engineering work and supporting the team through cross-cutting bug fixes and production issues

Senior Software Developer

Arctic AI

Feb 2022 – Sept 2024

Remote

- Developed RAG-based AI systems using LangChain and LLMs across production workflows, tuning chunking and retrieval strategy against domain-specific query patterns
- Implemented vector search infrastructure for semantic retrieval, improving answer grounding and reducing token spend by narrowing context to relevant passages
- Built an enterprise data platform on Azure (Databricks, Data Factory, Data Lake), consolidating overlapping pipelines and enabling scalable analytics across multiple client products
- Scaled data pipelines on Azure Databricks to support multi-source ingestion and advanced analytics
- Led a full platform redesign in Next.js, improving user engagement and site performance
- Implemented CI/CD pipelines with GitHub Actions and Docker, standardizing deployments across services
- Built a financial data processing tool on GCP that replaced manual reconciliation workflows
- Mentored engineers and contributed to system design and technical direction across projects

Software Developer

Neo Financial

Jun 2021 – Aug 2021

Remote

- Developed a system for designing and generating credit statements for printing at Symcor, using Node, React, AWS, Terraform, TypeScript, GraphQL, MongoDB, and Jest
- Engineered the decider microservice to route declined transactions to a third-party fraud detection API and persist responses, and optimized the bank microservice powering MasterCard payment gateway integration

Data Scientist (M.Eng Research)

University of Waterloo

Sept 2019 – Dec 2020

Waterloo, ON

- Built image classification models using transfer learning (Inception-ResNet-V2, EfficientNet), achieving 92.18% accuracy across 120 classes
- Improved pedestrian detection performance by re-annotating datasets and training Faster R-CNN models with optimized data pipelines

PROJECTS

Natural-Language Business Intelligence Platform

Python, FastAPI, LangGraph, Next.js

- Built a multi-tenant BI platform using FastAPI, LangGraph, LiteLLM, and Next.js, where an LLM agent translates plain-English questions into SQL across Postgres, MySQL, SQLite, and Snowflake, selects an appropriate visualization, and assembles the result into shareable dashboards
- Extended the platform with a semantic business-term layer, natural-language alert definitions, automated anomaly and root-cause analysis, and usage-based billing via Stripe, with Celery and Redis handling scheduled and long-running analysis jobs

On-Device Voice Dictation (macOS, Android)

Rust, Swift, Kotlin

- Built a push-to-talk dictation tool that transcribes speech directly to the cursor in any application, running speech recognition and text cleanup entirely on device so no audio leaves the machine at runtime
- Implemented a single Rust core (whisper.cpp transcription, Silero voice activity detection, session state machine) exposed through UniFFI bindings to native shells, a Swift and AppKit menu-bar app on macOS and a Kotlin input method on Android

AI-Assisted Strength Training App

React Native, TypeScript, FastAPI

- Built an offline-first mobile app using React Native, Expo, TypeScript, SQLite, Supabase, and FastAPI that converts a conversational intake into a personalized multi-week training program and auto-progresses working weights from logged set performance

- Constrained LLM calls to the edges of a deterministic training engine, validating every model output against program invariants before it can modify a plan, keeping generation flexible while making progression logic reproducible

EDUCATION

M.Eng, Electrical and Computer Engineering (Distinction), University of Waterloo Sept 2019 – Dec 2020

B.Eng, Electrical and Electronics Engineering (First Class), Federal University Oye Feb 2013 – Jan 2018